

Fair and Constraint-Aware Group Formation

An interpretable decision-support prototype

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Abstract

We design and implement a decision-support system that forms fair and balanced groups in collaborative settings such as Erasmus+ youth exchanges. The task is modelled as a constraint-based optimization problem: participants are partitioned into fixed-size groups (hard constraints) so as to minimize a weighted combination of soft penalties capturing skill balance, diversity, and preference satisfaction. We implement and compare four methods — a random baseline, swap-based local search, simulated annealing, and an integer-linear-programming (ILP) formulation — evaluate them with fairness metrics that are deliberately independent of the optimization objective, analyse robustness under participant addition/removal, and provide faithful per-assignment explanations. On a synthetic cohort of 60 participants, the optimized methods roughly halve the cost of the random baseline; warm-started re-solving keeps assignments stable under perturbation; and a Pareto analysis makes the skill-versus-preference trade-off explicit. We close with an ethical discussion of fairness, inclusion, and transparency, framing the tool as support for—not a replacement of—human judgement.

1 Introduction

Group composition strongly shapes inclusion, participation, and learning outcomes in collaborative programmes. In contexts such as Erasmus+ youth exchanges, organizers must split heterogeneous participants into balanced teams, weighing competing goals: groups should be comparable in skill, diverse in background, and respectful of stated preferences. Done by hand, this is slow and opaque; done naïvely (e.g. at random), it is fast but frequently unfair. Framed computationally, this is a constraint-based partitioning problem closely related to the *maximally diverse grouping problem* [1] and to team-formation tasks [2].

This project asks how such decisions can be *partially automated* while remaining fair, robust, and interpretable. We contribute (i) a formal constraint-based model of fair group formation; (ii) four solvers spanning heuristic and exact paradigms; (iii) an evaluation protocol with metrics decoupled from the objective; (iv) a robustness study; (v) a lightweight explainability layer; and (vi) an ethical analysis. The system is a Python prototype driven from the command line, producing the figures reported here.

2 Problem Modelling

Let $P = \{1, \dots, n\}$ be the participants and $\mathcal{G} = \{g_1, \dots, g_K\}$ the groups. An *assignment* is a function $\pi : P \rightarrow \mathcal{G}$.

Hard constraints. Every participant belongs to exactly one group, and each group g has a fixed capacity c_g :

$$\sum_{g \in \mathcal{G}} \mathbf{1}[\pi(i) = g] = 1 \quad \forall i \in P, \quad \sum_{i \in P} \mathbf{1}[\pi(i) = g] = c_g \quad \forall g \in \mathcal{G}. \quad (1)$$

Capacities form a balanced partition of n (sizes differ by at most one). These constraints are inviolable; an assignment satisfying them is *feasible*.

Soft constraints. Each soft goal is expressed as a *penalty* in $[0, 1]$ (lower is better), so they can be combined linearly.

Each participant i has a skill vector $s_i \in [0, 1]^d$. Let $m_g = \frac{1}{|g|} \sum_{i \in g} s_i$ be the per-group skill mean. The *skill-balance* penalty is the across-group variance of these means, averaged over skills and normalized by the maximal variance 0.25 of values in $[0, 1]$:

$$\Phi_{\text{skill}}(\pi) = \frac{1}{d} \sum_{j=1}^d \frac{\text{Var}_{g \in \mathcal{G}}(m_{g,j})}{0.25}. \quad (2)$$

For each categorical attribute a (e.g. nationality, gender) with global distribution q_a and per-group distribution $p_{g,a}$, the *diversity* penalty is the mean total variation distance to the global mix:

$$\Phi_{\text{div}}(\pi) = \frac{1}{|A| K} \sum_{a \in A} \sum_{g \in \mathcal{G}} \frac{1}{2} \sum_c |p_{g,a}(c) - q_a(c)|. \quad (3)$$

Let R be the set of directed preference relations: a *like* ($i \rightarrow j$) is satisfied when $\pi(i) = \pi(j)$, a *dislike* when $\pi(i) \neq \pi(j)$. The *preference* penalty is the unsatisfied fraction:

$$\Phi_{\text{pref}}(\pi) = 1 - \frac{|\{r \in R : r \text{ satisfied}\}|}{|R|}. \quad (4)$$

Objective. The optimizers minimize the weighted cost

$$C(\pi) = w_{\text{skill}} \Phi_{\text{skill}}(\pi) + w_{\text{div}} \Phi_{\text{div}}(\pi) + w_{\text{pref}} \Phi_{\text{pref}}(\pi), \quad (5)$$

with weights w encoding the organizer’s priorities. Unless stated otherwise we use $w_{\text{skill}} = 1$, $w_{\text{div}} = 1$, $w_{\text{pref}} = 0.5$.

3 Fairness Metrics

To avoid circular evaluation, results are judged with metrics computed *independently* of C : **skill imbalance** (across-group variance of group skill means, unnormalized), **experience imbalance** (variance of per-group mean experience level), **diversity imbalance** (mean total-variation distance, as above), and **preference satisfaction** (the honored fraction $= 1 - \Phi_{\text{pref}}$). The first three are “lower is fairer”; the last is “higher is better”. This separation lets a method that games the objective be exposed by the metrics.

4 Algorithms

We implement four solvers spanning a random baseline, local-search metaheuristics [3, 4], and exact constraint programming [5].

Random baseline. A seeded uniform feasible assignment, used as the reference point.

Local search (hill climbing). Starting from a feasible random assignment, we repeatedly propose swapping two participants in different groups. Swaps preserve group sizes, so feasibility is invariant. A swap is accepted iff it strictly lowers C ; the search stops at an iteration cap or after a no-improvement plateau. Crucially, the cost change of a swap is evaluated *incrementally*: only the two affected groups’ skill sums and category counts and the preference relations incident to the swapped pair are recomputed, giving $O(\text{changed})$ per step rather than $O(n)$. This incremental evaluator is verified against full recomputation in the test suite.

Simulated annealing (SA). Identical machinery, but a worsening swap of cost change $\Delta > 0$ is accepted with probability $\exp(-\Delta/T)$, where the temperature T is cooled geometrically [4]. SA can escape local optima and returns the best assignment seen.

Integer linear programming (ILP). We encode binary variables $x_{i,g} \in \{0, 1\}$ with the hard constraints of Section 2 and solve with OR-Tools CP-SAT [6]. The objective is a *linear surrogate* of (5): the preference term is exact (same-group indicators via the linearization $z = x_{i,g} \wedge x_{j,g}$), the diversity term is exact up to integer rounding of targets, but the skill term is an *approximation* — variance is not linear, so we minimize the L_1 deviation of each group’s skill total from its balanced target. This is a faithful surrogate for *balance* but a different norm than the variance used by the metrics; we return to its consequences in Section 6.

5 Experimental Setup

We generate a synthetic cohort of $n = 60$ participants (seed 42, **skewed-skill** scenario: a minority of high performers among mostly lower-skilled participants), with $d = 4$ skills, five ordinal experience levels, two diversity attributes (nationality, gender), and optional likes/dislikes. Groups have capacity 6 ($K = 10$). Each method is run across 5 seeds; ILP is given a 10 s time limit. All artifacts are regenerated by `scripts/run_analysis.py`.

6 Results

6.1 Method comparison

Table 1 reports means over 5 seeds; Figure 1 visualizes them. The optimized methods roughly halve the cost of the random baseline. SA achieves the lowest cost (0.342), with local search close behind (0.355) at a fraction of the runtime. ILP attains by far the best *skill* balance (0.0008 vs 0.003–0.007), exactly as its L_1 surrogate intends, but spends that advantage elsewhere: its preference satisfaction is the lowest of the optimized methods, so its overall cost is higher. This is the surrogate-versus-metric tension anticipated in Section 4, and a useful reminder that “optimal” is only optimal *with respect to the encoded objective*.

Table 1: Mean results over 5 seeds ($n = 60$, groups of 6). Lower is better except preference satisfaction.

Method	Cost	Skill imb.	Exp. imb.	Div. imb.	Pref. sat.	Runtime (s)
SA	0.3422	0.0029	0.1878	0.0900	0.5190	0.56
Local search	0.3551	0.0035	0.1844	0.0958	0.5095	0.19
ILP	0.4215	0.0008	0.1500	0.1110	0.3857	10.04
Random	0.7084	0.0073	0.3022	0.2830	0.2071	0.00

A subtle but important observation in Figure 1: the random baseline has the *lowest* experience imbalance. Because experience is not in the objective, the optimizers freely worsen it to improve skill, diversity, and preference. This is not a bug but a direct consequence of *what we chose to optimize* — and a concrete example of why the metric set is broader than the objective.

6.2 Convergence and skill distribution

Figure 2 shows cost decreasing monotonically for hill climbing and non-monotonically (with accepted uphill moves) for SA, which reaches a slightly lower final cost. Figure 3 shows the spread of per-group mean skill: tight for the optimized methods (balanced groups) and wide for random.

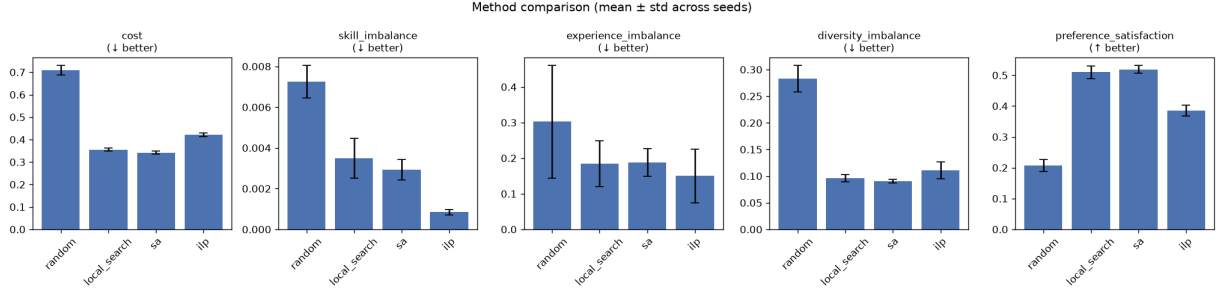


Figure 1: Per-metric comparison (mean $\pm$ std across seeds). Note that random achieves the *best* experience imbalance: experience is not part of the cost (5), so the optimizers trade it away to improve the weighted objectives.

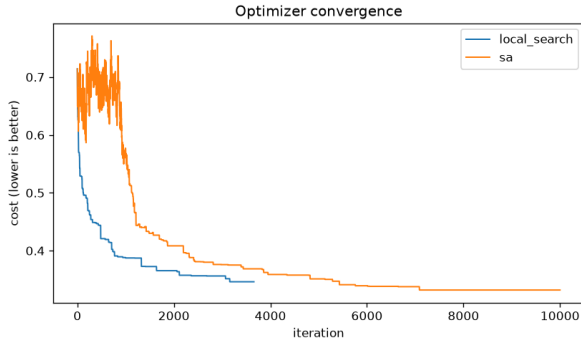


Figure 2: Cost vs iteration.

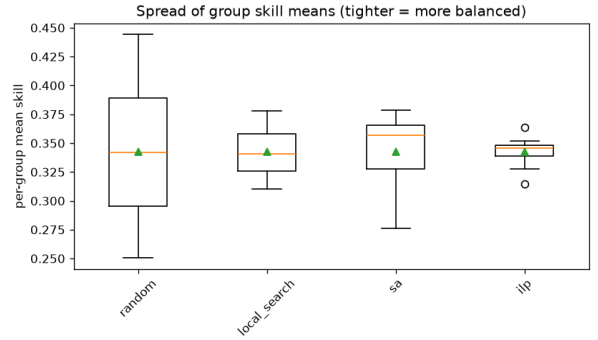


Figure 3: Per-group mean skill spread.

6.3 Robustness

We perturb the cohort by removing or adding $k \in \{1, 3, 6\}$ participants and re-solve with local search, comparing *cold* starts (fresh random initialization) against *warm* starts (re-optimizing from the previous solution, repaired to the new capacities). Stability is measured by the **group-change rate** (fraction of common participants who change group, after aligning arbitrary group labels by maximum overlap) and the label-invariant **pair-preservation** rate (fraction of participant pairs whose same-group status is preserved). Table 2 and Figure 4 summarize the study.

Table 2: Mean stability over 5 trials (local search). Warm-starting drastically reduces churn while fairness barely drifts.

Perturb.	Start	k	Group-change rate	Pair preservation
remove	cold	3	0.600	0.865
remove	warm	3	0.312	0.918
add	cold	3	0.570	0.866
add	warm	3	0.107	0.965

Warm-starting reduces the group-change rate by roughly 2–5 $\times$ relative to cold starts (e.g. 0.107 vs 0.570 for adding three participants) while keeping pair-preservation high. Across all settings the change in skill imbalance after perturbation is on the order of 10^{-3} , i.e. fairness is essentially preserved. For a real organizer, this means a late change to the roster need not reshuffle everyone: warm-started re-solving yields a stable, minimally disruptive update.

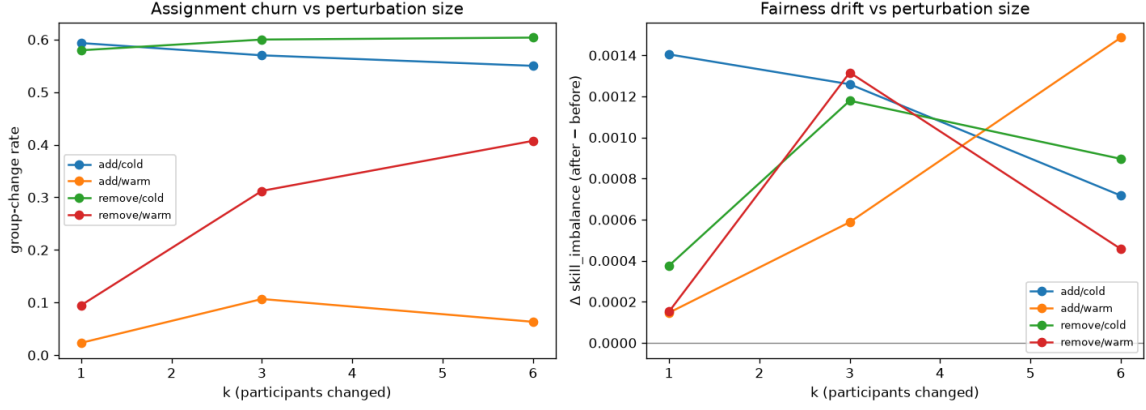


Figure 4: Assignment churn (left) and fairness drift (right) vs perturbation size k , for add/remove and cold/warm starts.

6.4 The fairness trade-off (Pareto front)

Soft objectives conflict: pushing skill balance can cost preference satisfaction. Sweeping the weight α between skill ($w_{\text{skill}} = \alpha$) and preference ($w_{\text{pref}} = 1 - \alpha$) traces the trade-off in Figure 5. The non-dominated (Pareto-optimal) settings lie at high α : at $\alpha = 1$ skill imbalance is minimal (0.00025) but preference satisfaction collapses to 0.27; at $\alpha = 0.8$ preference satisfaction peaks (0.64) at a small skill cost. Notably, for $\alpha \leq 0.7$ the solver converges to essentially the same solution — the trade-off only “bites” near the extremes. There is no single fair assignment: the choice along this curve is a *value judgement* that should rest with a human, which the system makes explicit rather than hiding inside a default weight.

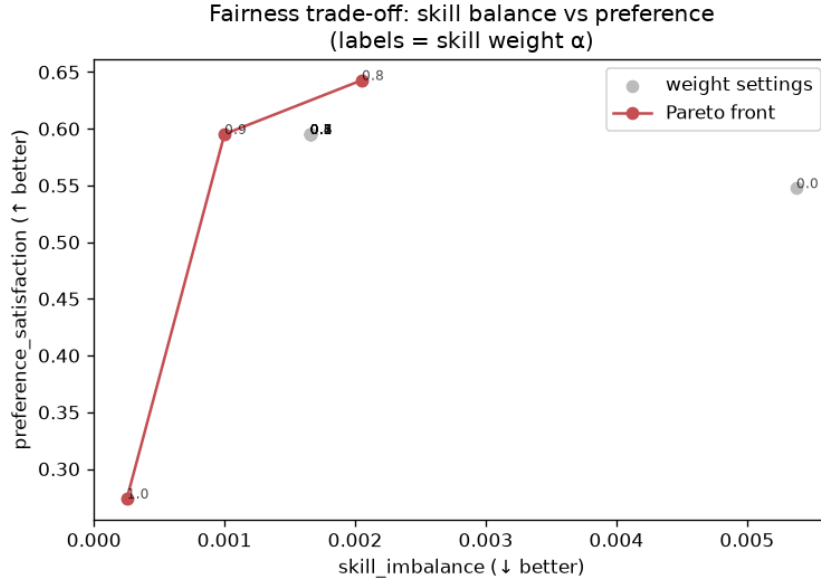


Figure 5: Skill balance vs preference satisfaction as the weight α varies. Labels show α ; the red curve is the Pareto front.

6.5 Explainability

For each participant the system produces a faithful explanation grounded in the cost model [11]. Because group sizes are fixed, the natural counterfactual is a *swap*: we report the single swap

that would most reduce cost (using the same incremental evaluator the optimizer uses), or state that the participant is “well-placed” if no swap improves the cost. On a converged local optimum, 86% of participants are well-placed. A concrete example:

Participant p007 → group g4

- Placement: could improve by swapping with p005 (group g2, $\Delta\text{cost} -0.0055$)
- Preferences: 1/2 satisfied --- grouped with liked p016; NOT grouped with liked p029
- Diversity: nationality=PL 33% in group vs 8% overall (over-represented)
- Skill: mean skill 0.27 lowers group mean 0.32 (overall 0.33)

Each factor is derived from the actual soft constraints, not reconstructed post-hoc by a separate surrogate model as in feature-attribution methods such as LIME [12], so the explanation is honest about *why* the placement was made and what its marginal effect is.

7 Ethical Discussion

Automating group formation touches directly on fairness, inclusion, and transparency [7, 10].

Fairness is plural and contested. Our system operationalizes fairness as a combination of skill balance, demographic representativeness, and preference satisfaction. These can conflict (Figure 5)—indeed, distinct fairness criteria can be provably impossible to satisfy simultaneously [9]—and there is no objective “most fair” weighting [8]. We therefore expose the weights as explicit, organizer-set parameters and visualize the trade-off, rather than encoding a hidden default that would silently impose one notion of fairness.

Inclusion. Diversity is measured against the *global* composition, which spreads minorities across groups instead of concentrating them. We deliberately tested a **minority-underrepresented** scenario; because the metric rewards matching the global mix, a rare group tends to be distributed rather than isolated. However, when a minority is extremely small, “even spreading” can also mean every such participant is the sole representative in their group — an outcome an organizer may or may not want. This is a value choice the tool should surface, not decide.

Transparency and contestability. Every assignment is accompanied by a faithful, human-readable explanation and a counterfactual (Section “Explainability”). This supports oversight: an organizer can see why someone was placed and what the cheapest alternative is, and can override it. Group labels are arbitrary, so we report label-invariant stability; small wording choices like this matter for not over-claiming.

Decision support, not automation. We frame the system as *support* for a human decision-maker. The optimizer encodes only what is written into the objective: experience balance, for instance, was *not* optimized and consequently degraded (Figure 1). This illustrates the central risk of such systems — they optimize precisely and only what they are told, including our blind spots — and argues for keeping a human in the loop who can notice unmodelled harms.

Limitations. (i) Data are synthetic; real attributes carry historical bias that a generator cannot faithfully reproduce, and naively learning “balance” from biased data could entrench it. (ii) The ILP objective is a linear *surrogate*; it is optimal for a proxy, not the true metric. (iii) Reducing people to skill vectors and categorical attributes is itself a reductive act with ethical weight. (iv) The preference model is simple (symmetric-ish likes/dislikes) and ignores power dynamics between participants.

8 Conclusion and Future Work

We presented an interpretable, constraint-aware prototype for fair group formation, compared heuristic and exact solvers, showed that optimization roughly halves the cost of random assignment while warm-started re-solving keeps assignments stable, and made the skill–preference trade-off explicit via a Pareto analysis. Throughout, the metric set is broader than the objective, which exposes exactly what the optimizer neglects. Future work includes: adding experience balance and group-level fairness floors as first-class objectives; multi-objective optimization that returns the whole Pareto set directly; richer, validated preference models; and a study with (consented, real) data to test whether the synthetic findings transfer.

Reproducibility. The full pipeline is regenerated by `python scripts/run_analysis.py`; individual steps are available as CLI commands (`generate`, `solve`, `evaluate`, `robustness`, `pareto`, `explain`). All results above were produced with seed 42.

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